

MINI PROJECT 6

ONLINE EXAM PORTAL

Test Plan

IEEE 829-style test planning document

Course	Software Engineering
Deliverable	2 - Test Plan
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Document Control

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Detailed Test Cases	Deliverable 4 - Test Case Update

1. Introduction

1.1 Purpose

This Test Plan defines the scope, approach, environment, responsibilities, entry/exit criteria, planned test scenarios, security validation, and requirement traceability for the Online Exam Portal. The plan is derived from the SRS and is aligned with the REST interfaces and component boundaries in the Software Architecture & Design Specification.

1.2 System Under Test

The system is a two-role web application. Admins create MCQ examinations, manage questions, view submissions, and release results. Students authenticate, attempt questions in any order, change answers, mark questions for review, submit manually or at timeout, and view results after release. The backend uses Python/Django with a relational database; browser clients communicate through HTTPS REST APIs.

1.3 References

- SRS - Online Exam Portal v1.0 (FR-01..FR-17, NFR-01..NFR-07, SEC-REQ-01..SEC-REQ-04).
- Software Architecture & Design Specification - Online Exam Portal v1.0.
- IEEE 829-style test-plan structure, used as academic guidance.
- Software Engineering Mini-Project Deliverables Part-1 brief.

2. Test Items

Test Item	Scope
Authentication	Student/Admin login, authenticated context, password storage expectations.
Exam Management	Create/read exams, availability window, four-option MCQ management.
Attempt Lifecycle	Start attempt, direct navigation, answer/change answer, mark for review, auto-save.
Submission & Grading	Manual submit, timeout auto-submit, score computation, attempt locking.
Results	Admin release control, Student result visibility, Admin submission list.
Quality Attributes	Performance, reliability, usability, availability, maintainability, scalability, portability.

Test Item	Scope
Security Controls	Authentication, RBAC, Student ownership, post-finalization integrity, HTTPS deployment.

3. Features to be Tested / Not Tested

3.1 In Scope

- All functional requirements FR-01 through FR-17 are within the overall plan.
- All non-functional requirements NFR-01 through NFR-07 are planned for validation using performance checks, walkthroughs, configuration review, and code/design inspection as appropriate.
- Security objectives SEC-OBJ-01 and SEC-OBJ-02, and security requirements SEC-REQ-01 through SEC-REQ-04, are explicitly validated.
- Student and Admin browser flows are checked together with API-level behavior so UI results and server-side rules remain consistent.

3.2 Out of Scope

- Proctoring and plagiarism detection.
- Non-MCQ question types.
- Self-registration and password-reset workflows.
- Load beyond the mini-project demonstration scale of 50 concurrent Students.

4. Test Approach

4.1 Test Types

Type	Method	Primary Coverage
Functional	Black-box API + UI workflow testing	FR-01..FR-17
Integration	Browser -> REST controller -> service -> ORM/database	Exam attempt, submit, grading, result release
Non-functional	Timed scripts, concurrency test, usability walkthrough, browser checks	NFR-01..NFR-07
Security	Negative authorization tests + configuration review	SEC-REQ-01..04, SEC-OBJ-01..02
Regression	Re-run affected and linked test cases after a defect fix	All changed flows

4.2 Test Environment

- Client: latest two versions of Chrome and Firefox on desktop/laptop.
- Server: Python 3.x, Django, standard Linux environment.
- Database: relational database (PostgreSQL preferred; SQLite acceptable for local demonstration).
- Network: standard broadband; deployed environment uses HTTPS.
- Test data: pre-provisioned Admin account, at least two Student accounts, representative exams with 5+ MCQs, and one short-duration exam for timeout testing.

4.3 Entry Criteria

- SRS and API endpoint definitions are baselined.
- Authentication, exam, attempt, submission, and result flows are implemented.
- Seed test accounts and exam data are available.

- Known blocking build/deployment issues are resolved.

4.4 Exit Criteria

- All planned critical/high-priority cases have executed.
- No unresolved defect prevents login, exam attempt, submission, grading, or result-release flows.
- Security validation has no unresolved unauthorized-access or post-submission-integrity failure.
- Any remaining non-blocking failures are documented with status and re-test decision.

4.5 Pass/Fail Rule

A test passes only when its observed result matches the stated expected result and quantitative threshold. A test fails when the behavior, state transition, response code, persisted data, timing threshold, or access-control result differs from the expected result.

5. Planned Test Case Set

The plan uses the following ten core cases for Part-1. Deliverable 4 contains the detailed preconditions, steps, inputs, expected results, and pass criteria for each case.

ID	Category	Scenario	SRS Trace
TC-01	Functional	Student and Admin login with valid credentials	FR-01, FR-02, SEC-REQ-01
TC-02	Functional	Answer in any order and change an answer	FR-06, FR-07
TC-03	Functional	Mark for Review and five-state question grid	FR-08, FR-09, FR-10
TC-04	Functional/Security	Manual submit, grading, and attempt lock	FR-11, FR-13, SEC-REQ-03
TC-05	Functional	Auto-submit at timeout and grading	FR-12, FR-13
TC-06	Functional/Security	Result hidden until Admin release	FR-14, FR-15, FR-16, SEC-REQ-02
TC-07	Non-functional	Question page performance with 50 Students	NFR-01, NFR-06
TC-08	Non-functional	Answer auto-save within 2 seconds	NFR-02
TC-09	Security	RBAC, authentication, and cross-account ownership	SEC-OBJ-01, SEC-REQ-01, SEC-REQ-02
TC-10	Security	Reject answer/submit changes after finalization	SEC-OBJ-02, SEC-REQ-03

5.1 Security Validation

Security Item	Validation Method	Evidence / Pass Condition
SEC-OBJ-01	TC-09 negative-access checks	Unauthenticated requests are denied; Student B cannot access Student A; Student cannot invoke Admin actions.
SEC-OBJ-02	TC-10 integrity checks	Finalized attempt rejects answer edits and duplicate/late submit; score remains unchanged.

Security Item	Validation Method	Evidence / Pass Condition
SEC-REQ-01	TC-01 + TC-09 + storage inspection	Every protected function requires authentication; password field is a salted one-way hash, never plaintext.
SEC-REQ-02	TC-06 + TC-09	RBAC and ownership rules enforced for Admin actions and Student attempts/results.
SEC-REQ-03	TC-04 + TC-10	Second submit and post-finalize modifications are rejected with conflict behavior.
SEC-REQ-04	Deployment configuration review	Application is served over HTTPS; plain HTTP is redirected or refused.

6. Requirement Traceability

SRS Requirement(s)	Planned Coverage
FR-01, FR-02	TC-01
FR-03, FR-04, FR-05	Planned Admin/exam-management functional checks in execution cycle
FR-06, FR-07	TC-02
FR-08, FR-09, FR-10	TC-03
FR-11, FR-13	TC-04
FR-12	TC-05
FR-14, FR-15, FR-16	TC-06
FR-17	Planned Admin submission-list check
NFR-01, NFR-06	TC-07
NFR-02	TC-08
NFR-03	5-user usability walkthrough
NFR-04	Availability-window observation / deployment review
NFR-05	Architecture/code inspection against layered design
NFR-07	Chrome/Firefox compatibility check
SEC-OBJ-01, SEC-REQ-01, SEC-REQ-02	TC-01, TC-06, TC-09
SEC-OBJ-02, SEC-REQ-03	TC-04, TC-10
SEC-REQ-04	HTTPS deployment configuration review

7. Risks, Assumptions and Test Deliverables

Item	Plan
Risk: timing tests vary by machine/network	Execute in a controlled environment and record baseline plus concurrent-load measurements.
Risk: API paths change during implementation	Treat the SADS endpoint catalogue as the baseline; reconcile test scripts before execution.
Assumption	Accounts are pre-provisioned and representative exam data can be seeded.
Deliverables	This Test Plan, execution evidence/defect log, and Deliverable 4 detailed test-case update.